Track 2: Visarjan Ghat Watch

Problem statement · Claude Impact Lab Bhopal · Sunday 11 October 2026 · SSR Global Skills Park

Turn three years of buried water tests into a warning Bhopal can act on before Durga visarjan.

Claude Impact Lab Bhopal was a one-day build event. Teams had from 10:15 AM to 5:00 PM to build a working tool for a real Bhopal problem with Claude and Claude Code. The projects are open source, and the work is offered to the Government of Madhya Pradesh through MPSeDC.

This document covers Track 2: the problem, the data given to teams, the rules of the day and how projects were judged.

Problem statement

Background. Every Ganesh and Durga festival, the MP Pollution Control Board (MPPCB) tests the water at Bhopal's idol-immersion ghats three times: before immersion, during, and one week after. The results go into reports of 90 to 120 pages. Nobody lines them up against the legal limits or compares one year with the next.

What the reports show. The numbers buried in those reports are alarming. At Bairagarh (Sant Hirdaram) kund, dissolved oxygen read 0 mg/l four times across 2023 and 2025. Water at zero oxygen supports no fish life at all, and the bathing-water standard asks for at least 5. Across all Bhopal ghat readings from 2023 to 2025, 42 of 116 dissolved-oxygen readings are below 5 mg/l, and 100 of 112 BOD readings are above the bathing limit of 3 mg/l.

Why now. The National Green Tribunal has banned immersion in the Upper and Lower Lakes, and Bhopal Municipal Corporation is building new ghats. Durga visarjan falls on 20 October 2026, nine days after the event.

Who it is for. The Bhopal Municipal Corporation officer who plans Durga visarjan at the ghats. This officer decides where to send extra cleaning, aeration and water testing. At present that decision rests on memory, because the test results sit in PDFs nobody reads.

Why it is hard.

- The reports print no units. The units in the data are the standard ones for each test.
- A blank means "not sampled", not "clean", and some ghats were sampled only during immersion.
- BDL means below the detection limit: very low, but not zero and not missing.
- Only 4 metals were tested (chromium, lead, zinc, copper).
- Two site names may each cover two places: "Rani Kamlapati ghat / Kamla Park kund" and "Hathaikheda dam / kund".

What teams were asked to build. A tool that gives the officer, before 20 October, a ranked list of Bhopal ghats by risk with the reason for each rank in plain words, the numbers behind every rank traceable to the report and page, and a short brief in Hindi and English the officer can forward. The tool reports the readings against the CPCB limits, says honestly where the data is missing, and claims no health effects the data doesn't show.

The data

Everything is in the `data/` and `resources/` folders of this track.

Files

File	Rows	What it holds
`data/bhopal_ghats_2023_2025.csv`	1,524	The main dataset. Every reading at 7 Bhopal ghats, 2023 to 2025, Ganesh and Durga, before / during / after. All 1,524 values match the PDFs
`data/sample_bhopal_2025.csv`	453	The 2025 rows of the file above, for quick testing
`data/mp_immersion_sites_2023_2025.csv`	21,600	The same tests at about 100 sites per festival across MP. Less checked. 879 rows are marked `needs_check` because the site name could not be read, and about 825 more have bad site names without the mark
`data/cpcb_water_quality_criteria.csv`	19	The official CPCB water-quality limits, classes A to E
`resources/`	6 PDFs	The original MPPCB reports (2023, 2024, 2025) and Upper / Lower Lake water-quality reports

Columns in the immersion CSVs

Column	Example	Meaning
`year`	`2025`	Report year
`festival`	`Ganesh` or `Durga`	Which immersion the test covers
`site`	`Bairagarh (Sant Hirdaram) kund`	One standard name per Bhopal ghat (Bhopal file only)
`water_body`	`Upper Lake`	Lake or kund the ghat sits on (Bhopal file only)
`regional_office`	`R.O. Bhopal`	MPPCB regional office (statewide file only)
`phase`	`pre`, `during`, `after`	Before immersion, during immersion, one week after
`parameter`	`Dissolved oxygen`	Standard parameter name
`value`	`2.5`	The number. Empty when the report shows no number
`unit_assumed`	`mg/l`	Unit. The reports print no units; these are the standard units for each test
`flag`	`BDL`, `not sampled`	Why `value` is empty
`value_as_printed`	`BDL`	Exactly what the report shows
`parameter_as_printed`	`D.Oxygen`	Parameter name exactly as printed
`site_as_printed`	`Sant Hirdaram Visarjankund, Bairagarh, Bhopal`	Site name exactly as printed
`site_match_confidence`	`high`, `medium`	How sure the standard name is (Bhopal file only)

Column	Example	Meaning
`needs_check`	`yes`	Site name could not be read (statewide file only)
`source_page`	`28`	PDF page in that year's report. Open the PDF and check

The 13 parameters

Parameter	Unit assumed	Plain meaning	Bathing limit (CPCB class B)
pH	none	Acid or alkaline. 7 is neutral	6.5 to 8.5
Dissolved oxygen	mg/l	Oxygen in the water. Fish need it	5 or more
BOD	mg/l	Biochemical oxygen demand: how much organic waste eats oxygen	3 or less
COD	mg/l	Chemical oxygen demand: total pollution that uses oxygen	none set
Total solids	mg/l	Everything left after the water dries	none set
Dissolved solids	mg/l	Solids dissolved in the water	none set
Suspended solids	mg/l	Particles floating in the water. Not in the Durga 2023 and Durga 2025 tables	none set
Turbidity	NTU	Cloudiness	none set
Conductivity	µS/cm	Dissolved salts. Class E caps it at 2,250	none set for B
Chromium, Lead, Zinc, Copper	mg/l	Heavy metals, often from idol paint	none set for B

Codes in the data

Code	Meaning	How to treat it
`BDL`	Below detection limit	Very low. Not zero, not missing
`ND`	Not detected	Very low
`not sampled`	Blank cell, `_` or `–` in the report	Missing. Never clean
`*`, `**`	Not sampled at that stage	Missing

The 7 Bhopal ghats

Standard name	Water body	Confidence	Note
Bairagarh (Sant Hirdaram) kund	Upper Lake	high	A temporary kund. Its conductivity is far above the Upper Lake's, so its readings are tank water, not lake water. No Durga 2025 table in the report
Khatlapura ghat	Lower Lake	high	
Prempura ghat	Upper Lake	high	
Rani Kamlapati ghat / Kamla Park kund	Lower Lake	medium	Ganesh tables say "Kamla Park kund", Durga tables say "Lower Lake ghat". May be two places

Standard name	Water body	Confidence	Note
Hathaikheda dam / kund	Hathaikheda dam	high	Ganesh tables say "visarjan kund", Durga tables say "dam (ghat)". May be two places
Malikhedi kund	Kund	high	No Durga 2025 table in the report
Shahpura lake ghat	Shahpura Lake	high	

CPCB water-quality criteria

From `data/cpcb_water_quality_criteria.csv`:

Class	Designated best use	Limits
A	Drinking water source without conventional treatment but after disinfection	Total coliform 50 MPN/100 ml or less · pH 6.5 to 8.5 · Dissolved oxygen 6 mg/l or more · BOD (5 days, 20°C) 2 mg/l or less
B	Outdoor bathing (organised)	Total coliform 500 MPN/100 ml or less · pH 6.5 to 8.5 · Dissolved oxygen 5 mg/l or more · BOD (5 days, 20°C) 3 mg/l or less
C	Drinking water source after conventional treatment and disinfection	Total coliform 5,000 MPN/100 ml or less · pH 6 to 9 · Dissolved oxygen 4 mg/l or more · BOD (5 days, 20°C) 3 mg/l or less
D	Propagation of wildlife and fisheries	pH 6.5 to 8.5 · Dissolved oxygen 4 mg/l or more · Free ammonia (as N) 1.2 mg/l or less
E	Irrigation, industrial cooling, controlled waste disposal	pH 6.0 to 8.5 · Electrical conductivity at 25°C 2,250 micromhos/cm or less · Sodium absorption ratio 26 or less · Boron 2 mg/l or less

Sources

Document	Link	Date
MPPCB Ganesh and Durga Visarjan Report 2025 (92 pages)	https://www.mppcb.mp.gov.in/proc/mon/Ganesh_and_Durga_visarjan_report-2025.pdf	Downloaded 10 Oct 2026
MPPCB Ganesh and Durga Visarjan Report 2024 (118 pages)	https://www.mppcb.mp.gov.in/proc/mon/Ganesh_and_Durga_visarjan_report-2024.pdf	Downloaded 10 Oct 2026
MPPCB Idol Immersion Report 2023, CPCB format (87 pages)	https://www.mppcb.mp.gov.in/proc/mon/Idol_immer._Report_23_CPCB.pdf	Downloaded 10 Oct 2026
Upper and Lower Lake monthly water quality 2025-26	https://xgn.mp.nic.in/File_Upload/MONITORING//Yearly%20Water%20Quality%20Status/2026/July/UPPER%20LAKE-2025-26..pdf	Downloaded 10 Oct 2026
Upper Lake water quality 2023-24	https://www.mppcb.mp.gov.in/proc/mon/Bhopal-2023-24_upper_lake.pdf	Downloaded 10 Oct 2026
Upper Lake water quality by station, 2003-04 to 2022-23	https://www.mppcb.mp.gov.in/proc/mon/comparatives-uppe-lak-lower-lake-2003-2023.pdf	Downloaded 10 Oct 2026
CPCB water quality criteria	https://cpcb.gov.in/water-quality-criteria/	Read 10 Oct 2026
NGT immersion ban and new BMC ghats, Free Press Journal, 26 Feb 2026	https://www.freepressjournal.in/amp/bhopal/mp-news-idol-immersion-to-shift-from-lakes-to-safer-sites-ngt-ban-pushes-bmc-to-build-4-dedicated-ghats	

Document	Link	Date
Durga visarjan date, 20 October 2026, Drik Panchang	https://www.drikpanchang.com/navratri/durga-puja/durga-visarjan-date-time.html?year=2026	

Hints given to teams

These were ideas, not instructions. Teams chose their own approach.

- Decide which CPCB class fits immersion ghats, and say why. Class B (outdoor bathing) is a reasonable start.
- Non-technical team members had real work here: researching the NGT order and CPCB guidelines, placing the ghats on a map, writing the Hindi brief and checking numbers against the PDFs.

Constraints

- Public data only. Everything teams needed was in this folder.
- No connection to any live government system.
- No health effects the data doesn't show. The tool reports the readings against the limits.

Rules of the day

1. **No pre-work.** Submitted code had to be written from 10:15 AM on the day. Libraries, frameworks and a team's own general tools were allowed.
2. **Any tools.** Teams could use Claude, Claude Code, Cerebras inference APIs or anything else that fit.
3. **The data in this pack.** Teams could add other public data, named and linked in their README.
4. **No live government systems.** No logging in to, scraping or writing to any government portal during the build.
5. **No personal data.** The data here has none.
6. **One workflow, one user, one outcome.** The tool was for the officer described above, not a portal.
7. **Teams of 4 to 5,** mixed technical and non-technical, with real work for everyone.
8. **Public repo by 5:00 PM.** The code is offered to the department, so it has to be readable.

How projects were judged

The jury brief called for the organisers plus a representative of the problem's real users: a department officer where possible, or a domain expert. Scoring leaned toward day-one impact, meaning what the user gets the week after the event.

Submissions were due at 5:00 PM. The jury and mentors then reviewed the submissions and picked the teams to present. Each presenting team had 6 minutes to present and demo, then 4 minutes of questions.

Each judge filled one card per presenting team. Every criterion was scored 0 to 5, and day-one impact counted double, so the maximum was 25.

Criterion	What is scored	Scoring guide
Day-one impact (×2)	Is it ready to use, with real impact, on day one?	5 ready, high impact · 4 almost ready, high impact · 3 ready, mild impact · 2 almost ready, mild impact · 1 not ready or not impactful
Product	User experience, ease of use, onboarding	5 polished, works first try · 4 well designed, fits the workflow · 3 usable with little learning · 2 functional, needs learning · 1 hard to use
Idea	How creative and impactful the idea is, whatever the result	5 rare insight · 4 creative and original · 3 solid but predictable · 2 close to what exists · 1 already exists
Clarity of demo	Are the problem, the user and the outcome obvious?	5 impeccable pitch and demo · 4 good pitch, convincing demo · 3 ok pitch, demo worked · 2 weak narrative · 1 no narrative or demo failed

For Track 2 the jury looked for:

- The ranking of ghats is traceable to the readings and the CPCB limits.
- Missing data shows as missing, never as clean.
- The brief is short and readable by a BMC officer in Hindi and English.
- The team is honest about what the data doesn't cover.

If two teams tied, the higher day-one impact score won.

What teams submitted

1. **A public GitHub repo link.** Required.
2. **A screen recording of 2 minutes or less.** Optional. A demo that failed live still counted if the recording worked.

Each repo had to contain:

Item	Why
`README.md` with team name, members, track, and the one user the tool is for	The jury read this first
How to run it, in 5 steps or fewer	The department must be able to run the work after the event
What works and what doesn't	Honesty scored better than a hidden gap
Any extra data used, with links	Rule 3
A `LICENSE` file	The code is offered to the department. MIT unless the organisers said otherwise
No API keys or secrets in the code	Keys belong in environment variables

After the event, the organisers connect winning teams with the relevant department officers so the work can move toward real use.

The projects submitted for this track are listed at https://github.com/Vishal4742/bhopal-impact-lab-2026-problem-statements#track-2-visarjan-ghat-watch